

The mode superimposition principle for distributed parameter systems

1 Introduction

While dealing with the study of lumped parameter multi-degree of freedom (d.o.f.) systems, one powerful approach is provided by the so called “mode superimposition principle” or “modal approach”: this is basically a particular choice of the set of independent coordinates, by which the motion of the system is described as a linear combination of motion components having the shape of the system’s modes of vibration. For multi-d.o.f. systems, the main advantage of this approach is that under appropriate assumptions concerning the damping matrix, the equations of motions are decoupled, so that the motion of a system having n d.o.f.s can be studied as the superimposition of n single-d.o.f. motion components, called “modal components”.

In this lesson, we introduce the mode superimposition principle or modal approach for a mono-dimensional distributed parameter system, like the tensioned string or beam studied in the previous lessons. To this aim, we consider the case of the bending vibrations for a pinned-pinned slender beam, but the same approach can be easily adapted to other boundary conditions and to the other 1-dimensional distributed parameter systems studied: transverse vibration of a tensioned string, axial vibration of a bar/beam and torsional vibration of a rod.

We consider therefore the system shown in Figure 1.1, also showing the sign conventions for the bending displacement w of the beam and for the internal shear force T and bending moment M .

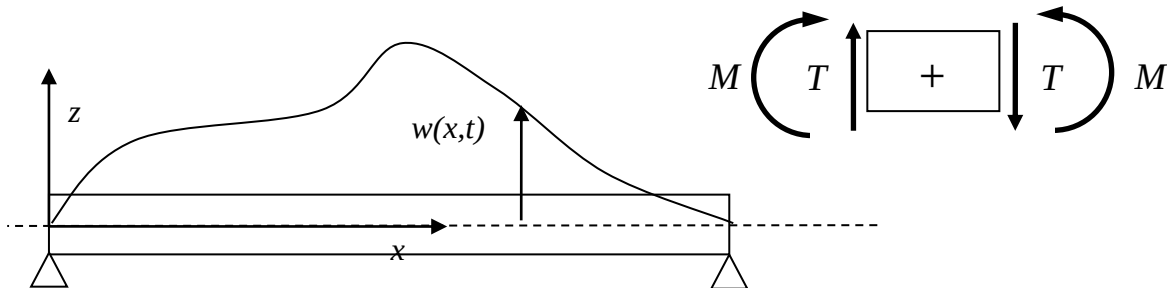


Figure 1.1: Pinned-pinned slender beam in bending.

The bending vibration of the beam, described by function $w(x, t)$, can be expressed as a linear combination of motion components having the shape of the modes of vibration according to the following series expansion:

$$w(x, t) = \sum_{k=1}^{\infty} \phi_k(x) q_k(t) \quad (1.1)$$

where $\phi_k(x)$ is the function defining the shape of the k -th mode of vibration, and $q_k(t)$ is the k -th modal coordinate introduced by Eq. (1.1). Consistently with the beam being a distributed parameter system (i.e. an “infinite-d.o.f.” system), an infinite number of modal components and coordinates appear in Eq. (1.1).

We note here that Eq. (1.1) represents the application of the mode superposition principle to any 1-dimensional distributed parameter system in general, provided of course that the appropriate mode shape functions $\phi_k(x)$ are used. In the particular case of the pinned-pinned beam in bending, we recall from Lesson 3 the expression of the mode shape functions:

$$\varphi_k(x) = \sin\left(\frac{k\pi x}{L}\right) \quad (1.2)$$

introducing these expressions in Eq. (1.1) leads to:

$$w(x,t) = \sum_{k=1}^{\infty} \sin\left(\frac{k\pi x}{L}\right) q_k(t) \quad (1.3)$$

which can be seen for any given value of time t as the expansion in a sine series of an odd function of space: this is a special case of Fourier series. therefore, in the specific case of the pinned-pinned beam, we recognise that Eq. (1.1) is not only justified by analogy with the mode superposition approach as introduced for a multi-d.o.f. system, but has a sound mathematical justification based on Fourier series. For other boundary conditions and for other types of 1-dimensional distributed parameter systems, an “expansion theorem” exists (see Meirovitch) providing theoretical ground for introducing a modal expansion of the solution in the form of Eq. (1.1).

In order to manage efficiently the mode superposition approach, we are interested in truncating the series in eq. (1.1) to a finite number of terms. to this aim, we approximate the solution $w(x,t)$ by the sum of a finite number of modal terms as follows:

$$w(x,t) = \sum_{k=1}^n \phi_k(x) q_k(t) \quad (1.4)$$

To ensure the sum in Eq. (1.4) is a good approximation of the solution, we choose the number n of terms in the sum so that:

$$\omega_{n+1} \gg \Omega_{MAX} \quad (1.5)$$

where ω_{n+1} is the lowest natural frequency of the system associated with a modal component not considered in the approximated solution defined by Eq. (1.4) and Ω_{MAX} is the maximum frequency possible for all forcing effects being applied to the system.

As we will show later, based on Eq. (1.4) the motion of the system can be described as the superimposition of n modal components whose importance in defining the overall motion of the system depends on the amplitude of the corresponding modal coordinate q . This amplitude, in turn, depend on the quasi-static displacement of the coordinate, which is inversely proportional to the stiffness term appearing in the 1-d.o.f. equation of motion, and on the dynamic amplification factor depending on the ratio of the forcing frequency over the 1-d.o.f. system's natural frequency.

If Eq. (1.5) is satisfied, then any term in the exact solution (1.1) which is neglected in the approximated solution (1.4) will provide a small / negligible contribution to the solution because:

- i) the corresponding stiffness term is very large compared to the terms considered by Eq. (1.4): it will be shown indeed that the modal stiffness is increasing with the 4th power of the order k of terms in Eq. (1.4);
- ii) the dynamic amplification factor is close to 1 and hence no important dynamic amplification effect takes place, because on account of Eq. (1.5) excitation takes place in the quasi-static frequency range for the modal component in question.

Finally, we may re-write Eq. (1.4) in matrix form as:

$$w(x,t) = \underline{\Phi}^T(x) \underline{q}(t) \quad (1.6)$$

with $\underline{\Phi}(x)$ the vector collecting the scalar. space dependent functions defining the shapes of the n modes of vibration considered by the approximated solution Eq. (1.4) and $\underline{q}(t)$ the vector collecting the n modal coordinated introduced by Eq. (1.4):

$$\underline{\Phi}(x) = \begin{Bmatrix} \phi_1(x) \\ \phi_2(x) \\ \dots \\ \phi_n(x) \end{Bmatrix} ; \quad \underline{q}(t) = \begin{Bmatrix} q_1(t) \\ q_2(t) \\ \dots \\ q_n(t) \end{Bmatrix} \quad (1.7)$$

We close this section with one important remark: according to the mode superposition approach, the independent coordinates of the system are the modal coordinates $q_k(t)$ $k=1, \dots, n$, which are finite in number. Therefore, we recognise that, by using the approximated solution defined by Eq. (1.4) or, equivalently Eq. (1.6), we approximate the motion of the distributed parameter system as that of an n-d.o.f. system. This kind of approximation goes under the name of *discretisation*, as it reduces a distributed parameter system to a concentrated parameters one.

2 Equations of motion

In order to write the equations of motion arising from the application of the mode supersposition principle to the slender beam in bending, we use Lagrange equation in matrix form:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}} \right)^T - \left(\frac{\partial T}{\partial q} \right)^T + \left(\frac{\partial D}{\partial \dot{q}} \right)^T + \left(\frac{\partial V}{\partial q} \right)^T = \bar{f}(t) \quad (2.1)$$

The expression of the kinetic energy E_c is:

$$T = \frac{1}{2} \int_0^L \left(\frac{\partial w}{\partial t} \right)^2 m dx = \frac{1}{2} \int_0^L \left(\frac{\partial w}{\partial t} \right)^T m \left(\frac{\partial w}{\partial t} \right) dx \quad (2.2)$$

Deriving Eq. (1.6) with respect to time we get:

$$\frac{\partial w}{\partial t} = \underline{\Phi}^T(x) \dot{\underline{q}}(t) \quad (2.3)$$

and bringing (2.3) in (2.2) we obtain:

$$T = \frac{1}{2} \dot{\underline{q}}^T \int_0^L \underline{\Phi}(x) m \underline{\Phi}^T(x) dx \dot{\underline{q}} = \frac{1}{2} \dot{\underline{q}}^T [\bar{m}] \dot{\underline{q}} \quad (2.4)$$

with:

$$[\bar{m}] = \int_0^L \underline{\Phi}(x) m \underline{\Phi}^T(x) dx \quad (2.5)$$

the modal mass matrix of the system (considering the first n modal components), which is a square, diagonal matrix of order n . Giving a proof of the mass matrix being diagonal is straightforward if we restrict ourselves to the case of the pinned-pinned beam. Indeed, recalling the expression of the modal shapes provided by Eq. (1.2), we get the following expression for the generic element m_{ij} in row i and column j of the matrix:

$$\bar{m}_{i,j} = m \int_0^L \sin\left(\frac{i\pi x}{L}\right) \sin\left(\frac{j\pi x}{L}\right) dx = \begin{cases} \frac{mL}{2} & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (2.6)$$

In case of different boundaries conditions or of other 1-dimensional vibration problems studied, an “orthogonality theorem” holds, see the book by Meirovitch, allowing to conclude that in all cases the mass matrix generated by the mode superposition is diagonal.

Next, we write the potential energy V for the system. To this aim, we note first that if the system is moving in a vertical plane, the potential energy is the sum of two terms, one accounting for elastic forces generated in the beam and the other one for gravitational forces. However, we recall from concentrated parameters systems that if the height of the centre of gravity of all masses in the system depends linearly on the independent coordinates of the system, then by choosing as the zero of the independent coordinates the static equilibrium position under the effect of gravitational forces, the elastic potential energy associated with the elastic static loads in the equilibrium position cancels with the gravitational potential energy, so that the total potential energy can be written as if the system was moving in a horizontal plane (i.e not subject to the effect of gravitational forces).

To write the elastic potential energy, we use the Clapeyron's theorem, stating that the elastic potential energy is one half of the work of the elastic forces times the elastic deformation of the system. For bending deformation, the non-zero components of the elastic internal forces generated are the shear force and the bending moment. However, for a slender beam we assume shear deformation to be negligible, so that the elastic potential energy is only related to the work of the bending moment. Thus, the infinitesimal contribution dV to the potential energy given by an infinitesimal piece of beam having length dx is:

$$dV = \frac{1}{2} M d\vartheta \quad (2.7)$$

where M is the bending moment and $d\vartheta$ is the relative rotation between the two faces of the infinitesimal piece of beam considered. Now:

$$d\vartheta = \vartheta(x+dx) - \vartheta(x) = \frac{\partial \vartheta}{\partial x} dx$$

and, for a slender beam, the rotation of sections are obtained as the gradient of the bending displacement w (see lesson 3)

$$\vartheta = \frac{\partial w}{\partial x}$$

therefore:

$$d\vartheta = \frac{\partial^2 w}{\partial x^2} dx$$

replacing the above result in (2.7) and recalling from continuum mechanics that for the case we are treating:

$$M = EI \frac{\partial^2 w}{\partial x^2}$$

we obtain:

$$dV = \frac{1}{2} EI \left(\frac{\partial^2 w}{\partial x^2} \right)^2 dx = \frac{1}{2} EI \left(\frac{\partial^2 w}{\partial x^2} \right)^T \left(\frac{\partial^2 w}{\partial x^2} \right) dx$$

and the total potential energy for the beam is:

$$V = \frac{1}{2} \int_0^L EI \left(\frac{\partial^2 w}{\partial x^2} \right)^T \left(\frac{\partial^2 w}{\partial x^2} \right) dx \quad (2.8)$$

From (1.6) we get:

$$\frac{\partial^2 w}{\partial x^2} \cong \underline{\Phi}^{''T}(x) \underline{q}(t) \quad (2.9)$$

and substituting in (2.8) we obtain:

$$V = \frac{1}{2} \underline{q}^T \int_0^L \underline{\Phi}''(x) EI \underline{\Phi}''^T(x) dx \underline{q} = \frac{1}{2} \underline{q}^T [\bar{k}] \underline{q} \quad (2.10)$$

with:

$$[\bar{k}] = \int_0^L \underline{\Phi}''(x) EI \underline{\Phi}''^T(x) dx \quad (2.11)$$

Matrix $[\bar{k}]$ is the modal stiffness matrix of the system (considering the first n modal components), which is also a square, diagonal matrix of order n . As for the mass, matrix, it is easy to show that the stiffness matrix is diagonal for the pinned-pinned boundary condition case (for other cases, the orthogonality theorem holds, see again the book by Meirovitch).

Recalling again the expression of the modal shapes Eq. (1.2), the expression of the generic element k_{ij} in row i and column j of the stiffness matrix is:

$$\begin{aligned} \bar{k}_{ij} &= EJ \int_0^L \varphi''_i(x) \varphi''_j(x) dx = EJ \int_0^L \left(-\left(\frac{i\pi}{L}\right)^2 \sin\left(\frac{i\pi x}{L}\right) \right) \left(-\left(\frac{j\pi}{L}\right)^2 \sin\left(\frac{j\pi x}{L}\right) \right) dx = \\ &= EJ \left(\frac{i\pi}{L}\right)^4 \frac{L}{2} \quad \text{if } i = j \\ &= 0 \quad \text{if } i \neq j \end{aligned} \quad (2.12)$$

We note here that the diagonal term in row i of the stiffness matrix, i.e. the modal stiffness parameter associated with the i -th mode of vibration, is increasing with the 4th order of the mode, this confirms statement i) at page 2.

We also note that if we consider the case of the un-damped and free (not subject to time dependent forces) beam, then the applying Lagrange's equations in matrix form (2.1) we get the following set of equations of motions:

$$[\bar{m}] \ddot{\underline{q}} + [\bar{k}] \underline{q} = \underline{0} \quad (2.13)$$

and because the mass and stiffness matrices are diagonal, the natural frequencies are obtained by taking the square root s of the ratios of the modal stiffness parameters over the modal masses which in turn, considering the expression of the modal stiffness parameters from (2.13) and of the modal masses from (2.6) gives the first n natural frequencies of the pinned-pinned beam in bending as obtained in Lesson 3:

$$\omega_i = \sqrt{\frac{\bar{k}_{ii}}{\bar{m}_{ii}}} = \left(\frac{i\pi}{L}\right)^2 \sqrt{\frac{EJ}{m}} = \quad (2.14)$$

hence, we see that the modal superposition approach preserves the first n natural frequencies of the un-damped system.

Using the mode superposition principle, we can account in a simplified way for dissipative effects taking place in the vibration of the distributed parameter system. In most cases, these dissipative effects are mainly related to imperfect elastic behaviour of the material (known as material hysteresis) and / or to interaction of the vibrating system with air, so that a (normally small) portion of the kinetic energy of the system is transformed into kinetic energy of the neighbouring air and then in heat through dissipative effect related to air viscosity.

Defining a model for these effects based on the physical processes related to energy dissipation would be too complicated and is out of scope in this course. Generally, the approach which is followed is based on defining an ‘equivalent’ viscous damping producing comparable dissipative effects in the system. To this end, we define a viscous dissipative function as:

$$D = \frac{1}{2} \dot{\underline{q}}^T [\underline{c}] \dot{\underline{q}} \quad (2.15)$$

Then, we assume the damping matrix $[\underline{r}]$ to be diagonal and we choose the diagonal terms of the matrix so as to match the non-dimensional damping factors¹ h_i^s of the system obtained from experiments (cf. next lessons on identification of modal parameters). Therefore, we set as the modal damping parameters r_{ii} for the system the following values:

$$\bar{c}_{ii} = h_i^s c_{ci} = h_i^s 2m_{ii}\omega_i = h_i^s 2\sqrt{k_{ii}m_{ii}} \quad (2.16)$$

The final step to be addressed in order to write the equations of motion of the system is to derive the expression of the Lagrangian components $\underline{Q}(t)$ of the time-dependent forces acting on the system.

We consider here two notable cases: a single concentrated force and a distributed, time dependent load. A third case, consisting of one or more moving loads, is treated separately in Section 3.

For a single, time varying force $F(t)$ applied in a fixed position along the beam, see figure 2.1, the expression of the virtual work is:

$$\delta L = \delta w(\bar{x}) F(t) = \delta w(\bar{x})^T F(t) \quad (2.17)$$

with $\bar{x}$ the abscissa corresponding to the position where the load is applies, cf. Figure 2.1.

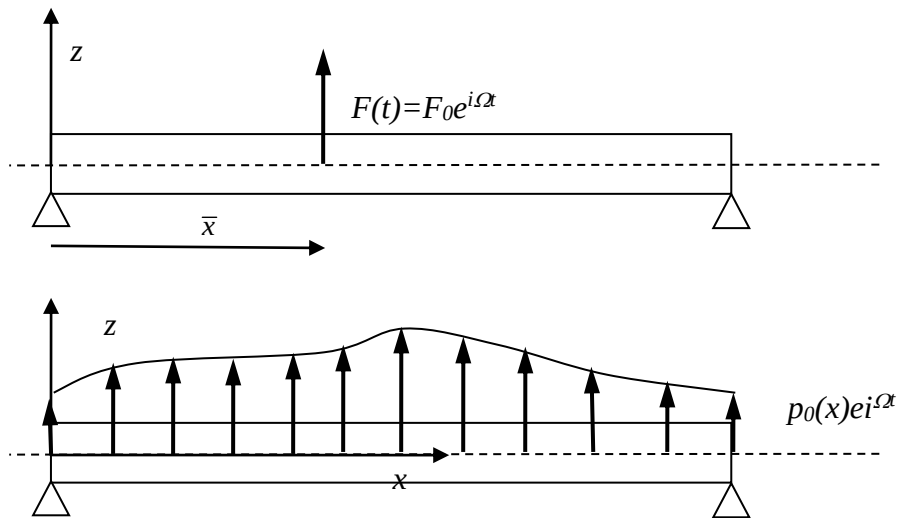


Figure 2.1: Virtual work for a single concentrated force (above) and for a distributed load (below).

From (1.6) we get:

$$\delta L(x) = \underline{\Phi}^T(x) \delta \underline{q} \quad (2.18)$$

¹ We recall that for a 1-d.o.f. system, the non-dimensional damping factor h is the ratio of the damping parameter r over the critical damping r_c , i.e. the minimum amount of damping that causes the free motion of the system to be non-oscillatory:

$$h = \left(\frac{c}{c_c} \right)$$

and, replacing (2.18) in (2.17):

$$\delta L = \delta \underline{q}^T \underline{\Phi}(\bar{x}) F(t) = \delta \underline{q}^T \underline{\bar{f}}(t) \quad (2.19)$$

with:

$$\underline{\bar{f}}(t) = \underline{\Phi}(\bar{x}) F(t) \quad (2.20)$$

Based on (2.20), the Lagrangian component of a single concentrated force along each modal coordinate q_i is hence the product of the force times the corresponding modal shape evaluated at the section where the force is applied. Therefore, if the force is applied in a nodal section for the i -th mode, the Lagrangian component $Q_i(t)$ will be identically zero, and that modal component will not contribute to the forced vibration of the system (except for the initial transient phase affected by the initial condition applied on the system, cf. Lesson 1).

We now consider the case of a distributed load acting on the beam, as shown in the lower part of Figure 2.1. For the sake of simplicity, we assume the distributed load to vary harmonically with time, whereas we consider a general expression for the variation of the load with the space coordinate x . Therefore, the distributed load is defined by the function $p(x,t) = p_0(x)e^{i\Omega t}$.

Considering an infinitesimal piece of beam having length dx , the virtual work $d\delta L$ associated with the portion of the load applied on the infinitesimal piece of beam is:

$$d\delta L = \delta w(x) p_0(x) e^{i\Omega t} dx = \delta w(x)^T p_0(x) e^{i\Omega t} dx$$

and the virtual work associated with the entire distributed load is obtained by means of integration along the whole length of the beam:

$$d\delta L = \int_0^l \delta w(x)^T p_0(x) dx e^{i\Omega t} \quad (2.21)$$

Introducing (2.18) in (2.21) we get:

$$d\delta L = \delta \underline{q}^T \int_0^l \underline{\Phi}(x) p_0(x) dx e^{i\Omega t} = \delta \underline{q}^T \underline{\bar{f}}(t) \quad (2.22)$$

with

$$\underline{\bar{f}}(t) = \int_0^l \underline{\Phi}(x) p_0(x) dx e^{i\Omega t} = \underline{f}_0 e^{i\Omega t} \quad (2.23)$$

We recall from Lesson 3 that if the vibrating system is symmetric, as in the case of the pinned-pinned beam, then the shape of the modes of vibration will be either symmetric or anti-symmetric. Then, if a spatially symmetric load distribution function $p_0(x)$ is applied on the symmetric system, all the Lagrangian components related with anti-symmetric modes of vibration ($i=2, 4, 6, \dots$) will be zero and these modal components will not contribute to the forced vibration of the system, except for the initial transient phase affected by the initial condition. Conversely, if the symmetric system is forced by an anti-symmetric load distribution all the Lagrangian components related with symmetric modes of vibration ($i=1, 3, 5, \dots$) will be zero, so these modal components will not contribute to the stationary forced vibration of the system.

In conclusion, the following expressions were found for the kinetic and potential energy, dissipation function and virtual work, based on the mode superposition approach:

$$T = \frac{1}{2} \dot{\underline{q}}^T [\bar{m}] \dot{\underline{q}}; V = \frac{1}{2} \underline{q}^T [\bar{k}] \underline{q}; D = \frac{1}{2} \dot{\underline{q}}^T [\bar{c}] \dot{\underline{q}}; \delta L = \delta \underline{q}^T \underline{\bar{f}}(t) \quad (2.24)$$

Applying Lagrange equations in matrix form (2.1) to expressions (2.24) the following set of equations of motions in matrix form is obtained:

$$[\bar{m}]\ddot{\underline{q}} + [\bar{c}]\dot{\underline{q}} + [\bar{k}]\underline{q} = \underline{\bar{f}}(t) \quad (2.25)$$

and because the mass, damping and stiffness matrices are all diagonal, this equation reduces to n decoupled single-d.o.f. equations of motion in the form:

$$\begin{aligned} \bar{m}_{11}\ddot{q}_1 + \bar{c}_{11}\dot{q}_1 + \bar{k}_{11}q_1 &= \bar{f}_1(t) \\ \dots \\ \bar{m}_{ii}\ddot{q}_i + \bar{c}_{ii}\dot{q}_i + \bar{k}_{ii}q_i &= \bar{f}_i(t) \\ \dots \\ \bar{m}_{nn}\ddot{q}_n + \bar{c}_{nn}\dot{q}_n + \bar{k}_{nn}q_n &= \bar{f}_n(t) \end{aligned} \quad (2.25b)$$

Equations (2.25b) can be solved using the methods known for single-DOF systems.